

TRAN NGUYEN DUY BAO

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LinkedIn — GitHub — Portfolio

Professional Summary

AI Engineer with 3 years of production experience building end-to-end LLM systems, RAG pipelines, and multi-agent applications in healthcare and media. Owns full lifecycle from data ingestion and indexing through retrieval optimization, evaluation, and deployment on GPU clusters. Combines deep understanding of retrieval architectures (hybrid search, reranking, multimodal ingestion) with MLOps discipline to ship reliable, enterprise-grade AI systems.

Technical Skills

Core: RAG Architectures, Multi-Agent Systems, LLM Fine-Tuning, Retrieval Optimization, Prompt Engineering
Frameworks: LangChain, LangGraph, LlamaIndex, vLLM, PyTorch, Hugging Face Transformers, LangFuse, Camel-AI
Data & ML: Dagster, PySpark, Kafka, RabbitMQ, XGBoost, Scikit-learn, SpaCy, YOLOv8
MLOps: LangFuse, TensorBoard, Triton Inference Server, Docker, Git, GCP, AWS, NVIDIA H100
Backend & DBs: FastAPI, Flask, PostgreSQL, MongoDB, Pinecone, Elasticsearch
Languages: Python, SQL, JavaScript

Professional Experience

Trivita AI

Ho Chi Minh City, Vietnam

AI Engineer

Oct. 2025 – Present

- Optimized inference throughput for **MoE LLMs up to 397B parameters** (Qwen3, MedGemma) on 8xH100 cluster via tensor/expert parallelism, KV cache tuning, and quantization with vLLM; profiled and tuned serving parameters (batch size, prefill scheduling, speculative decoding) to meet sub-second latency SLAs.
- Deployed Sortformer (non-autoregressive ASR) to production via **NVIDIA Triton Inference Server** with dynamic batching, co-serving embedding and classification models alongside LLM endpoints on a shared GPU pool.
- Built a two-tier **LLM-as-a-Judge evaluation system** combining heuristic pre-filtering with **Prometheus** evaluators to score medical dialogues on hallucination rate, clinical accuracy, and treatment compliance.
- Curated **10–15K multi-turn preference pairs** for **RLHF/DPO** alignment of medical dialogue models; constructed chosen/rejected conversation pairs from clinician annotations using **Unsloth** for memory-efficient supervised fine-tuning and DPO training on consumer-grade GPUs.
- Fine-tuned **HuggingFace LLM model** for Vietnamese ICD-10 code classification on a heavily imbalanced long-tail dataset; applied augmentation and resampling across **~30 diagnostic codes** to recover minority class recall.
- Monitored latency, retrieval quality (Recall@k, MRR), and prompt efficiency via **LangFuse** and **TensorBoard**, feeding regression data into iterative prompt refinements that reduced per-query token consumption.

MedVita – Synthetic Medical Conversation Generation Pipeline

2025 – Present

- Designed a **Dagster orchestration pipeline** automating end-to-end generation and evaluation of synthetic doctor-patient conversations using a strict 4-stage clean architecture: preprocessing, inference, postprocessing, and evaluation.
- Architected multi-agent systems with **LangChain** and **Camel-AI** to simulate realistic clinical dialogues across 10+ specialized medical departments, with dynamic patient persona generation and scenario synthesis.

MCV Complex

Ho Chi Minh City, Vietnam

AI Engineer

Jul. 2023 – Oct. 2025

- Shipped 4 production AI systems across healthcare and media verticals, owning end-to-end delivery from data ingestion and RAG pipeline design through model selection, evaluation, and deployment on GCP and AWS.
- Introduced a modular inference engine toggling between local open-source models (vLLM, Ollama) and cloud APIs, routing low-complexity queries to local models to reduce API spend.

SpaceOne – Real-Time Media Summarization & Sentiment Analysis

2023 – 2025

- Designed a two-module pipeline: isolated vocals with **Spleeter**, transcribed with **Whisper**, then summarized via structured **GPT-4** prompts over LangChain; integrated into a **Kafka** streaming architecture for real-time processing at scale.
- Fine-tuned **PhoBERT** on domain-specific Vietnamese social media comments for 3-class sentiment classification, reaching **F1-score of 0.84** on real-world data.

Doctor Assistant – Multimodal Medical Chatbot

2023 – 2025

- Built a **LangGraph-powered** conversational agent with persistent memory and 6 integrated tools, orchestrating multi-turn clinical dialogue across text, PDF, speech (Whisper), and image inputs with dynamic tool selection.
- Architected end-to-end RAG pipeline: ingested **~2000 medical PDFs** and **~1,000 web sources**, chunked with metadata enrichment, and indexed into **Pinecone** with hybrid search (dense + BM25); added reranking and relevance thresholding to reduce hallucinated responses in clinical context.

Spoofage – FaceID with Anti-Spoofing Detection

2024

- Fine-tuned **YOLOv8** to classify real vs. spoofed inputs (printed images, screens, masks) on a custom dataset with synthetic and real-world scenarios; integrated **DeepFace** for post-liveness identity verification.
- Achieved **92% spoof detection accuracy** with sub-1s latency in live event conditions, reducing unauthorized check-ins by over **85%**.

Diabetes Risk Prediction from Tabular Survey Data

2025

- Trained 10+ regression models on OCR-parsed survey data with log-transformed glucose targets, ADASYN over-sampling, and RobustScaler normalization; SVR achieved $R^2 > 0.85$ with **~5% mean prediction error**.

Open Source

Meddies-OCR – Co-First Author

2025 – present

- Designed and trained a lightweight **VLM for medical OCR** by fine-tuning **LFM-2.5 VL** using **LoRA/PEFT** with supervised fine-tuning (SFT) and GOLD knowledge distillation, transferring OCR reasoning from a Chandra2 teacher to a 450M-parameter LFM-2.5 VL student model.
- Built the full training pipeline with **TRL** and **Unsloth** for memory-efficient LoRA training; optimized for low-latency inference on constrained hardware without sacrificing task-specific accuracy.

Education

Ton Duc Thang University

Ho Chi Minh City, Vietnam

B.Sc., Computer Science

Graduated Jun. 2025

GPA: 8.01 / 10.0 — Top 10, Recursion Hackathon 2022 — English Proficiency: C1

- Capstone: Fine-tuned **YOLOv8** on NIH ChestX-ray14 with CLAHE and elastic distortion augmentation for radiology abnormality detection; achieved **70.2% mAP** and **60.4% recall** on held-out test set.

References

Available upon request:

Dr. Nguyen Trang Thao — Former PhD at TDTU Lab — thao.nguyentrang@vlu.edu.vn

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